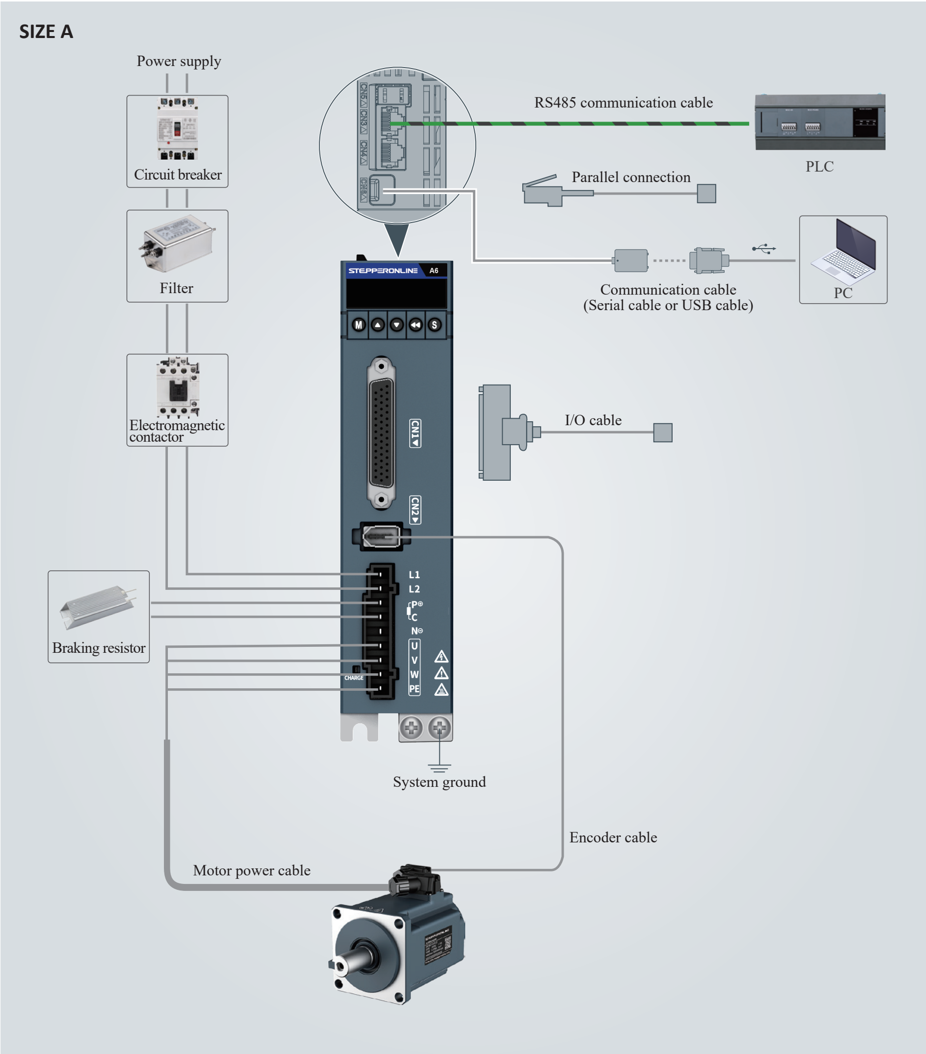
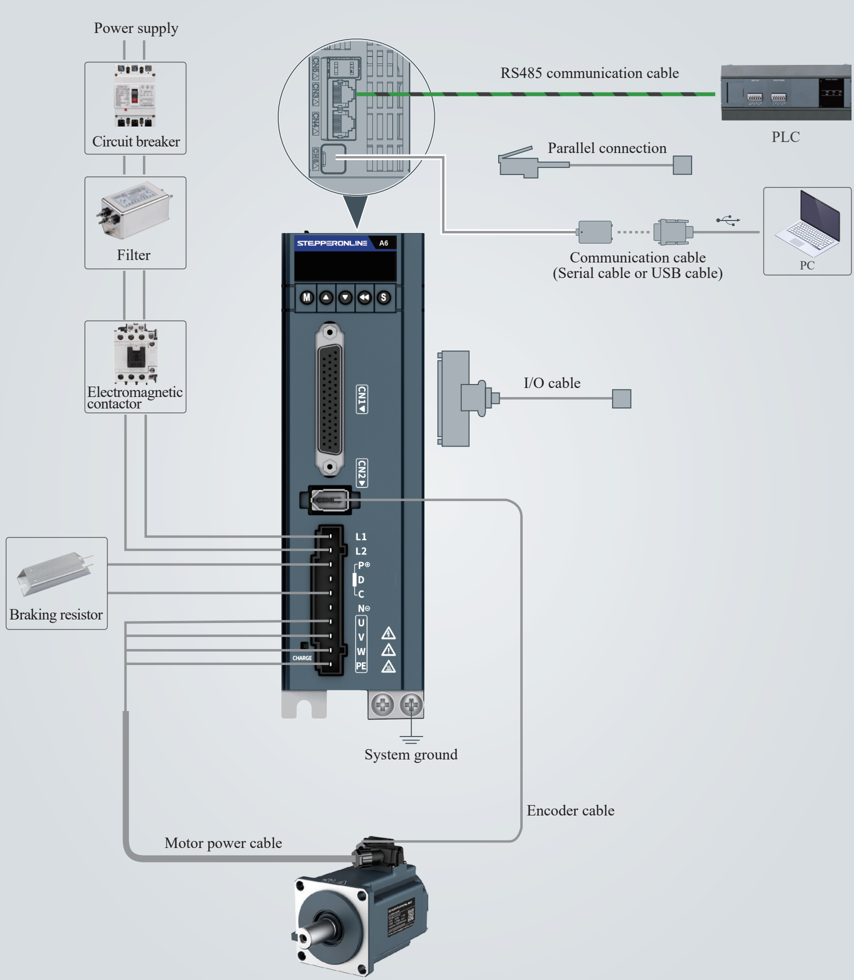
Chapter 3 Electrical Installation

3.1 System Topology



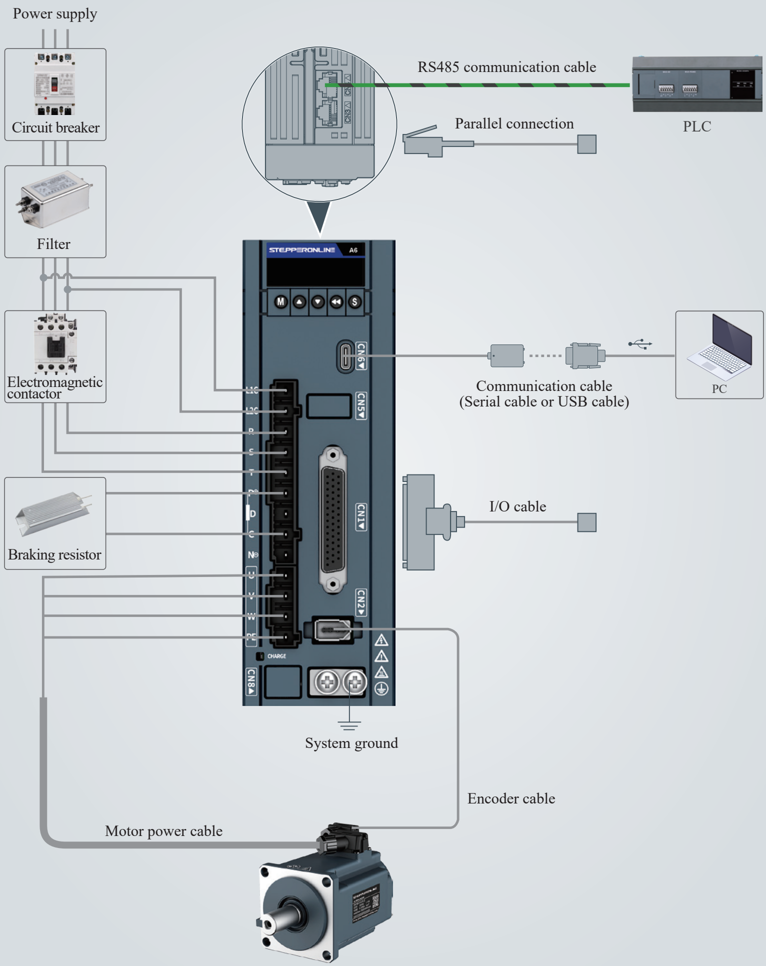
SIZE B



NOTICE

- Remove the jumper bar between terminals P⊕ and D before connecting an external braking resistor.

SIZE C



NOTICE

- Remove the jumper bar between terminals P⊕ and D before connecting an external braking resistor.

3.2 System Wiring

WARNING

- Only electrical engineering specialists can perform wiring.
- Connect an electromagnetic contactor between the input power supply and the main circuit of the drive, to form a structure that can cut off the power supply on the power side of the drive. If no electromagnetic contactor is connected, continuous large current upon drive faults may cause a fire.
- Ensure that the input voltage of the drive is within the allowable range. Failure to comply may result in product faults.
- Connect the drive protective earth (PE) terminal to that of the control cabinet. Failure to comply may result in an electric shock.
- Insulate the connection part of power supply terminals during wiring of the power supply and main circuit. Failure to comply may result in an electric shock.
- Ground the entire system. Failure to comply may result in malfunction.
- After power-off, wait at least 10 minutes before further wiring operations because residual voltage exists after power-off. Failure to comply may result in an electric shock.

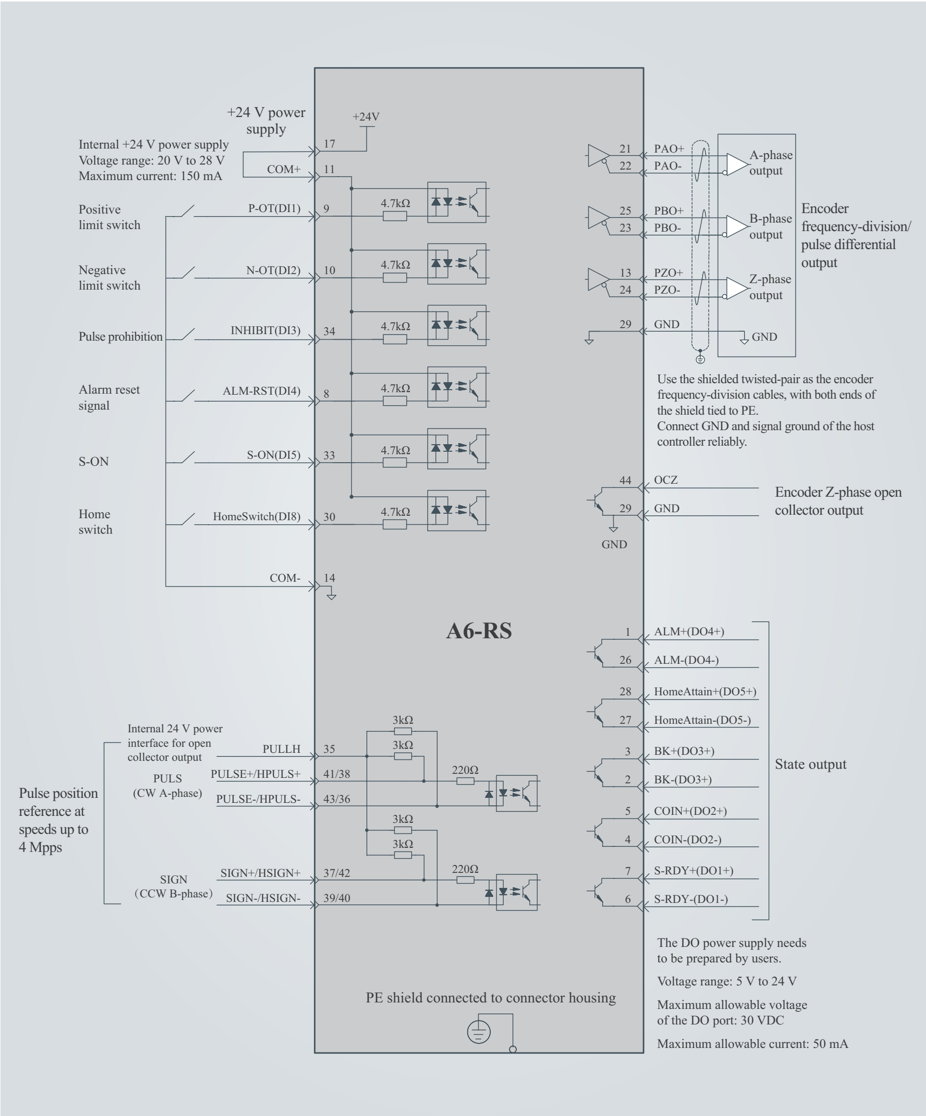
- Never power the servo drive with the IT grid. Use the TN or TT grid instead. Failure to comply may result in an electrical shock.
- Do not connect the output terminals U, V, and W of the drive to a three-phase power supply. Failure to comply may result in physical injury or a fire.
- Do not connect the motor terminals U, V, and W to a mains power supply. Failure to comply may result in physical injury or a fire.
- Do not power on the device before wiring is completed. Failure to comply may result in an electrical shock.

CAUTION

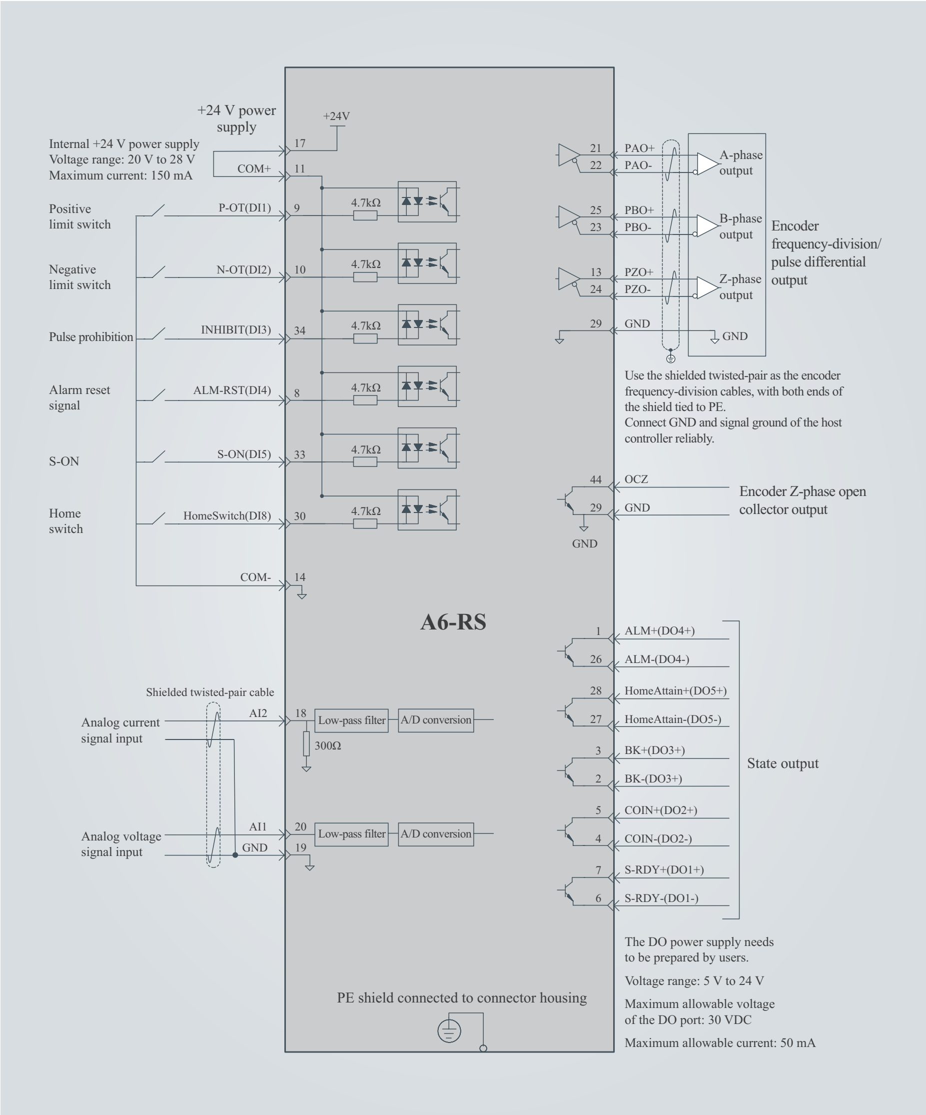
- Protect external wiring, branches, and short circuits according to local regulations.
- When using peripheral devices, read the user guide for each component and use it properly after fully confirming the precautions.
- Route the device properly. Improper wiring may cause damage to the drive and motor.
- Connect the drive to the motor directly, without connecting any electromagnetic contactor between them. Failure to comply may result in faults.
- Separate the main circuit cables from the I/O signal cables and encoder cables by at least 30 cm. Failure to comply may result in drive malfunction.
- Use twisted pairs or multi-conductor shielded twisted pairs as I/O signal cables or encoder cables. Failure to comply may result in drive malfunction.
- The maximum wiring lengths of I/O signal cables and encoder cables are 3 m and 10 m, respectively.
- Use a power filter to reduce electromagnetic interference on electronic devices around the drive.

- When wiring, do not allow conductive materials such as wire shavings to fall inside the drive.
- Never place cables under heavy objects or drag cables vigorously. Failure to comply may result in an electric shock due to cable damage.

Position mode



Torque mode


+24 V power supply
Internal +24 V power supply
Voltage range: 20 V to 28 V
Maximum current: 150 mA
+24V
17
COM+
11
Positive limit switch
P-OT(DI1)
9
Negative limit switch
N-OT(DI2)
10
Pulse prohibition
INHIBIT(DI3)
34
Alarm reset signal
ALM-RST(DI4)
8
S-ON
S-ON(DI5)
33
Home switch
HomeSwitch(DI8)
30
4.7kΩ
COM-
14
A6-RS
Shielded twisted-pair cable
Analog current signal input
AI2
18
Low-pass filter
A/D conversion
300Ω
Analog voltage signal input
AI1
20
GND
19
PE shield connected to connector housing
21
PAO+
22
PAO-
A-phase output
25
PBO+
23
PBO-
B-phase output
13
PZO+
24
PZO-
Z-phase output
29
GND
Encoder frequency-division/ pulse differential output
Use the shielded twisted-pair as the encoder frequency-division cables, with both ends of the shield tied to PE.
Connect GND and signal ground of the host controller reliably.
44
OCZ
29
GND
Encoder Z-phase open collector output
1
ALM+(DO4+)
26
ALM-(DO4-)
28
HomeAttain+(DO5+)
27
HomeAttain-(DO5-)
3
BK+(DO3+)
2
BK-(DO3+)
5
COIN+(DO2+)
4
COIN-(DO2-)
7
S-RDY+(DO1+)
6
S-RDY-(DO1-)
State output
The DO power supply needs to be prepared by users.
Voltage range: 5 V to 24 V
Maximum allowable voltage of the DO port: 30 VDC
Maximum allowable current: 50 mA

3.3 Ports

SIZE A

SIZE B

SIZE C

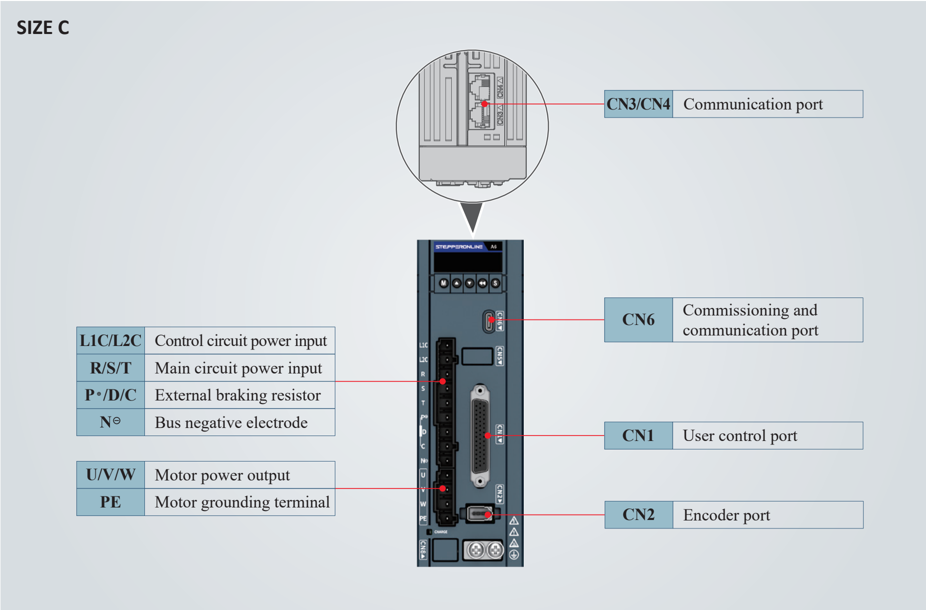
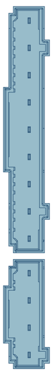
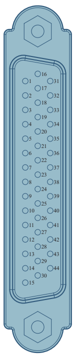
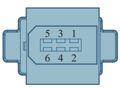
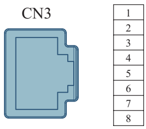
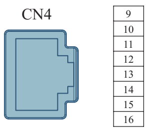
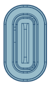


Table 3-1 Drive terminals

Terminal	Pin	Description
 Main circuit terminals	L1C and L2C: Control circuit power input terminals	Connected to the control circuit power supply as per the rated voltage class on the nameplate.
	R, S, and T: Main circuit power input terminals	Connected to the main circuit power supply as per the rated voltage class on the nameplate.
	P⊕ and N⊖: Servo bus terminals	Used when multiple servo drives share one DC bus.
	P⊕, D, and C: External braking resistor connection terminals	If an external braking resistor is needed, connect it between terminals P⊕ and C. ***Note:*** *Remove the jumper between terminals P⊕ and D before installing an external braking resistor. Otherwise, the braking transistor will be damaged due to overcurrent.*

Terminal	Pin		Description
	U, V, and W: Servo motor connection terminals		Connected to U, V, and W phases of the servo motor.
	PE: Motor grounding terminal		Connected to the grounding terminal of the motor for grounding purpose.
 CN1 user control terminal	7	DO1+	Servo ready
	6	DO1-	
	5	DO2+	Positioning completed
	4	DO2-	
	3	DO3+	Brake output
	2	DO3-	
	1	DO4+	Fault output
	26	DO4-	
	28	DO5+	Home attaining completed
	27	DO5-	
	9	DI1	Positive limit switch
	10	DI2	Negative limit switch
	34	DI3	Position reference inhibited
	8	DI4	ALM-RST (edge valid function)
	33	DI5	S-ON
	32	DI6	-
	12	DI7	-
	30	DI8	Home switch
	17	24V	Internal 24 V power supply Voltage range: 20 V to 28 V Max. output current: 150 mA
	14	COM-	
	11	COM+	Common terminal of DI terminals
	41	PULSE+	Low-speed pulse reference mode: • Differential drive input • Open collector Input pulse form: • Direction+pulse • Phase A + B quadrature pulse • CW/CCW pulse
	43	PULS-	
	37	SIGN+	
	39	SIGN-	

Terminal	Pin		Description
	38	HPULS+	High-speed input pulse reference
	36	HPULS-	
	42	HSIGN+	High-speed position reference symbol
	40	HSIGN-	
	35	PULLH	Input interface of external power supply for reference pulse
	21	PAO+	A-phase frequency-division output/fully closed-loop input
	22	PAO-	
	25	PBO+	B-phase frequency-division output/fully closed-loop input
	23	PBO-	
	13	PZO+	Z-phase frequency-division output/fully closed-loop input
	24	PZO-	
	29	GND	Signal ground
	44	OCZ	Encoder Z-phase open collector output
	15	5V	5 V power supply
	16	GND	Power ground
	20	AI1	Analog voltage signal input
	18	AI2	Analog current signal input
	31	AO1	Analog voltage output
	19	GND	Analog signal ground
 CN2 encoder terminal	1	+ 5V	5 V power supply
	2	0V	0 V power supply
	3	Reserved	-
	4	Reserved	-
	5	PS +	Encoder signal+
	6	PS -	Encoder signal-
	Enclosure	PE	Shield

Terminal	Pin		Description
 CN3 communication terminals	4	RS485+	Data transmit+
	5	RS485-	Data transmit–
	6	-	-
	7	-	-
	8	GND	Data receive–
	Enclosure	PE	Shielding layer
 CN4 communication terminals	12	RS485+	Data transmit+
	13	RS485-	Data transmit–
	14	-	-
	15	-	-
	16	GND	Data receive–
	Enclosure	PE	Shielding layer
 CN6 commissioning and communication terminal	Type-C		1: Type-c to serial, serial to USB 2: Type-c→USB

3.4 Power Supply Connection

CAUTION

- Do not connect the input power cables to the output terminals U, V and W. Failure to comply may cause damage the servo drive.
- Do not touch power terminals within 10 minutes after power-off because high voltage exists inside the drive after power off.
- Do not bundle power cables and signal cables together or route them through the same duct. Power cables and signal cables must be separated by a distance of at least 30 cm to prevent interference.
- Do not turn on and off the power supply frequently (once a minute). Failure to comply may cause drive faults.
- Do not power on the servo drive when any screw of the terminal block or any cable becomes loose. Failure to comply may cause a fire.

- Select cables based on the ambient temperature of the controller:
  When the ambient temperature is high, use heat-resistant cables. Teflon cables are recommended.
  Take heat preservation measures in low-temperature environments because the surface of regular cables may be easily hardened and cracked under low temperature.
- Use cables that can resist a voltage of 600 V AC or above and have a rated temperature of 75°C or above for the main circuit.
- Use a grounding cable with the same cross-sectional area as the main circuit cable. If the cross-sectional area of the main circuit cable is less than 1.6 mm^2, use a grounding cable with a cross-sectional area of 2.0 mm^2.
- Ground the drive reliably. Failure to comply may result in malfunction or even damage of the drive.
- Use shielded cables to comply with the EMC requirements.

3.4.1 Main circuit wiring

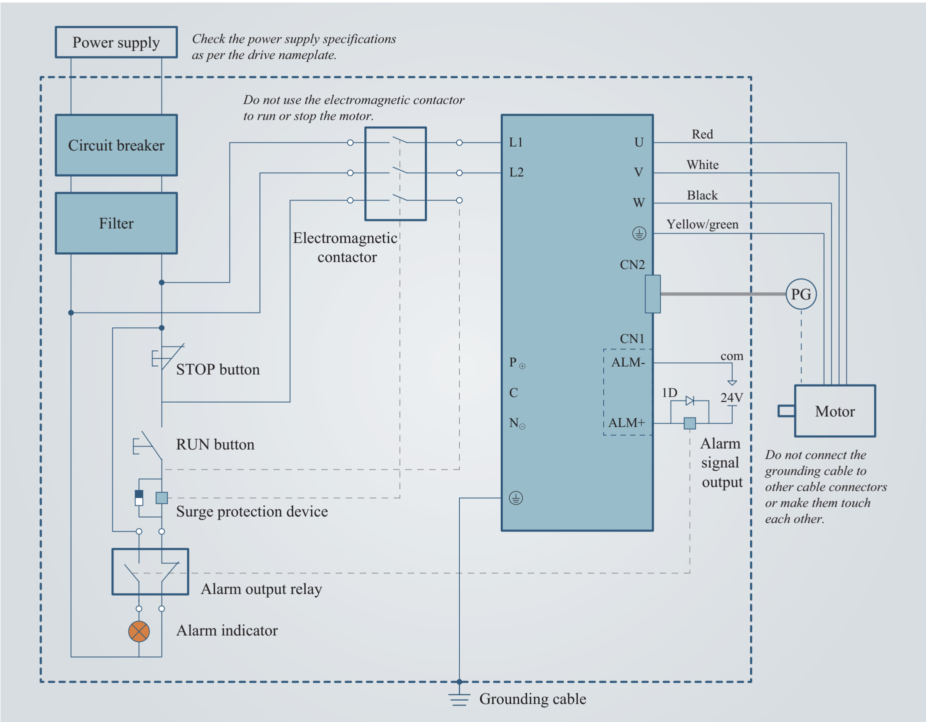


Figure 3-1 Wiring of the single-phase 220 V main circuit

NOTICE

Models with a single-phase 220 V power supply

- Model SIZE-A: A6-200RS, A6-400RS
- Model SIZE-B: A6-750RS
- Model SIZE-C: A6-1000RS(The main circuit can be connected to a single-phase or a three-phase 220 V power supply, depending on which one is available on site.)

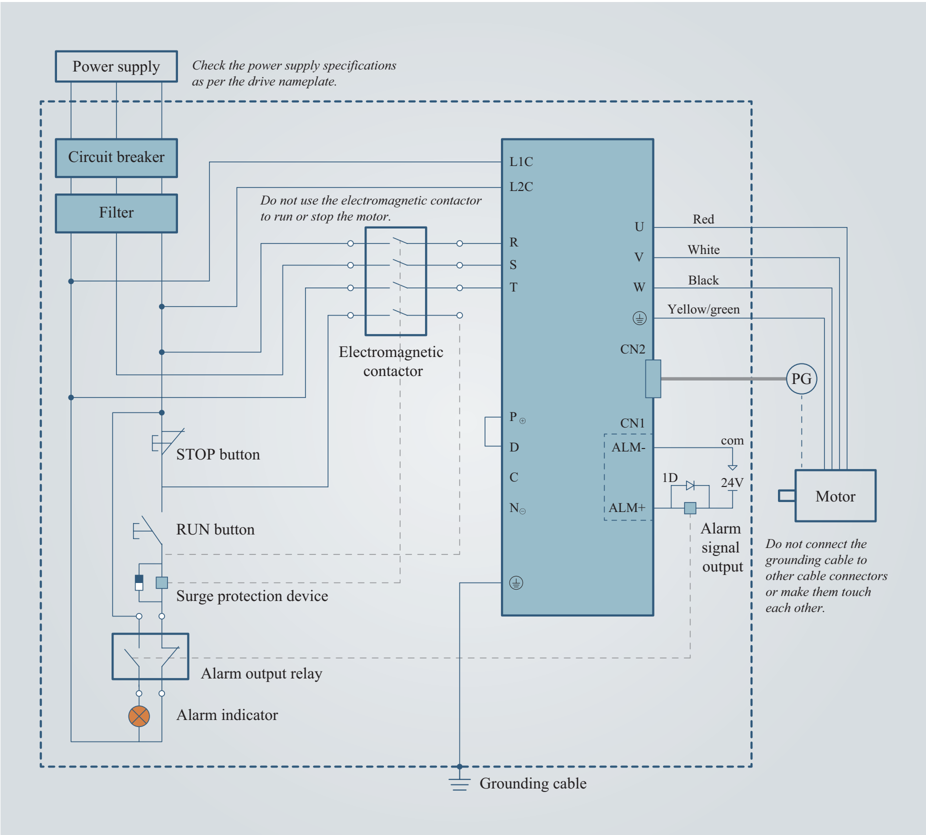


Figure 3-2 Wiring of the three-phase 220 V main circuit

NOTICE

Models with a three-phase 220 V power supply

- Model SIZE-C: A6-1000RS (The main circuit can be connected to a single-phase or a three-phase 220 V power supply, depending on which one is available on site.)

3.4.2 Cable specifications and recommendations

Table 3-2 Drive input and output current and recommended cables

Drive Model		Rated Input Current	Rated Output Current	Max. Output Current	Input Cable Specifications
Single-phase 220 V					
SIZE A	A6-200RS	2.3A	1.6A	5.8A	0.75 mm^2
SIZE A	A6-400RS	4 A	2.8 A	10.1 A	0.75 mm^2
SIZE B	A6-750RS	7.9 A	5.5 A	16.9 A	0.75 mm^2
SIZE C	A6-1000RS	9.6 A	7.6 A	23 A	1 mm^2
Three-phase 220 V					
SIZE C	A6-1000RS	5.1 A	7.6 A	23 A	0.75 mm^2

Table 3-3 Drive cable specifications and recommendations

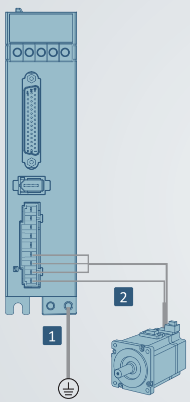
Cable Type	Cable Size	Outer Diameter (OD)
Power cable	4 × 12 AWG	12.2 ± 0.4 mm
	4 × 14 AWG	10.5 ± 0.3 mm
	4 × 16 AWG	9.5 ± 0.4 mm
	4 × 18 AWG	7.8 ± 0.2 mm
	4 × 20 AWG	6.5 ± 0.2 mm
Power shielded cable	4 × 12 AWG	12.9 ± 0.4 mm
	4 × 14 AWG	11.2 ± 0.4 mm
	4 × 16 AWG	10.1 ± 0.4 mm
	4 × 18AWG	8.3 ± 0.2 mm
	4 × 20 AWG	6.5 ± 0.2 mm
Power cable + brake cable	4 × 20 AWG + 2 × 24 AWG	6.5 ± 0.2 mm
Brake cable	2 × 18 AWG	5.8 ± 0.2 mm
	2 × 20 AWG	5.0 ± 0.2 mm

3.4.3 Grounding

CAUTION

- To prevent electric shock, ground the grounding terminal properly.
- Use grounding cables that meet technical standards for electrical devices and use protective grounding conductors that meet technical specifications. and shorten the grounding cable as much as possible.
- For use of multiple servo drives, ground them all. Improper grounding of the device may cause malfunction of the servo drive and the device.
- Do not use one grounding cable for multiple devices. Improper grounding of the device may result in servo drive or device faults caused by electrical interference.
- For drives equipped with selective grounding screws for VDR and insulation resistor, remove the selective grounding screw for VDR before voltage resistance test. Failure to comply may cause the servo drive to fail the test.
- Install the servo drive on a conductive metal mounting surface. Ensure that the whole conductive bottom of the device is attached properly to the mounting surface.
- Fix the grounding screw with the recommended torque. Avoid loosening or over-tightening of the protective grounding conductor.

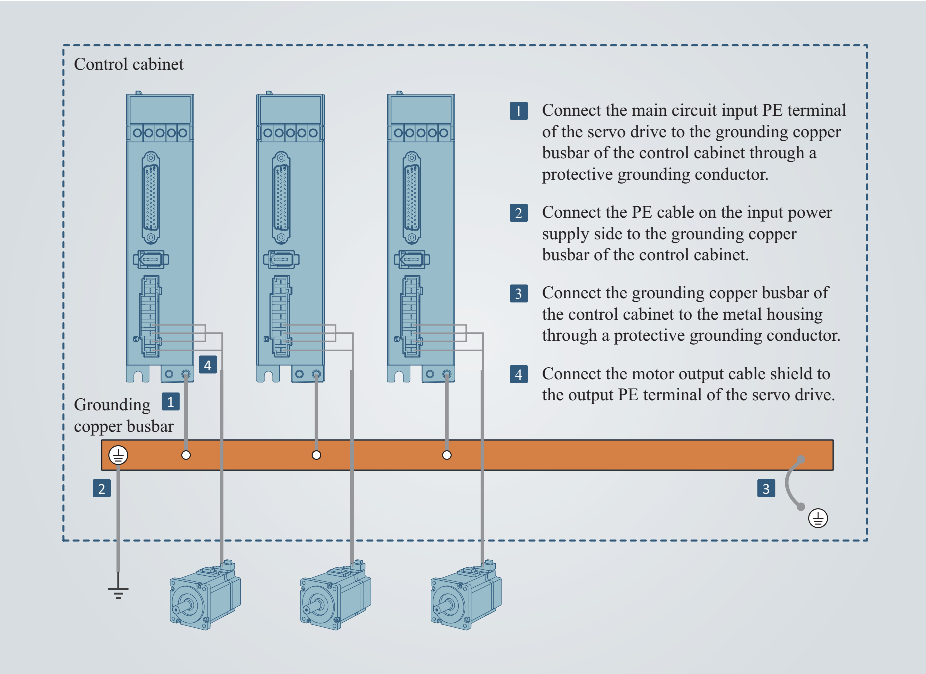
Grounding one servo drive alone



1 Connect the PE cable on the input power supply side to the input PE terminal of the drive.

2 Connect the output PE terminal of the drive to the motor output cable shield.

Grounding multiple servo drives



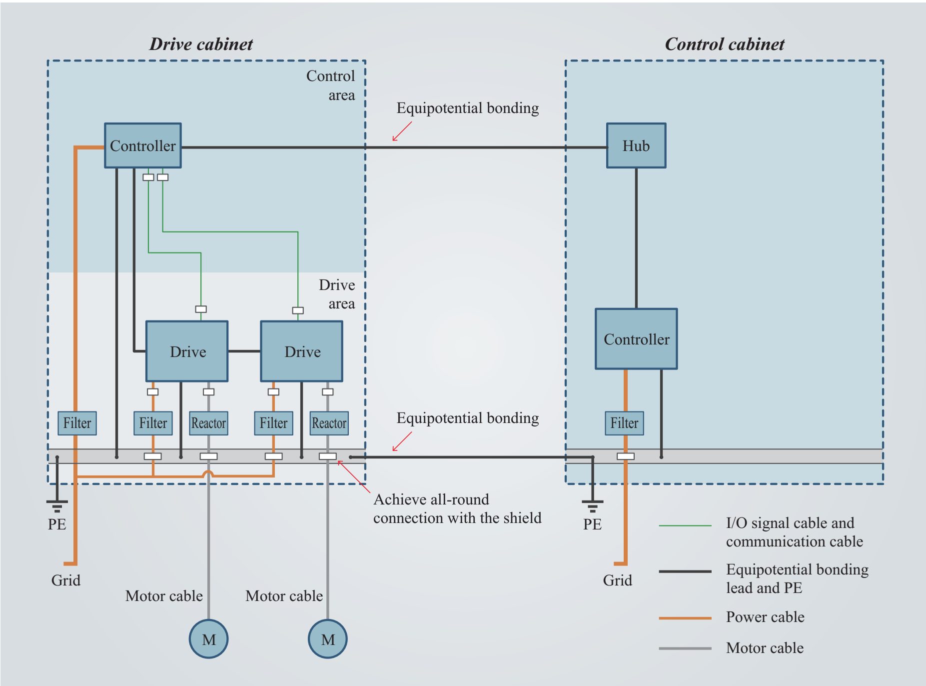
Grounding the control cabinet system

To suppress interference in the control cabinet, isolate the interference source from devices that may be interfered with. Divide the control cabinet into multiple EMC compartments or use multiple control cabinets based on the intensity of interference sources.

NOTICE

System installation principles:

- Place the control unit and the drive unit in two separate control cabinets.
- For installation involving multiple control cabinets, use a grounding cable with a cross-sectional area of at least 16 mm^2 to connect the control cabinets. This is to ensure equipotentiality between the cabinets.
- If only one control cabinet is used, place different devices in different compartments of the control cabinet based on signal intensity.
- Apply equipotential bonding to devices in different compartments inside the control cabinet.
- Shield all communication and signal cables drawn from the control cabinet.
- Place the power input filter in a position near the input interface of the control cabinet.
- Apply spray coating to each grounding point in the control cabinet.



Recommended grounding cable lugs for main circuit

Table 3-4 Recommended grounding cable lugs for power circuit

Drive Model		Rated Output Current	Lug Model of Power Cable	Lug Model of Brake Cable	Lug Model of PE Cable
SIZE A	A6-200RS	1.6A	E1008	E0508	TVR2-4
	A6-400RS	2.8 A	E1008	E0508	TVR2-4
SIZE B	A6-750RS	5.5 A	E1008	E0508	TVR2-4
SIZE C	A6-1000RS	7.6 A	E1508	E1008	TVR2-4

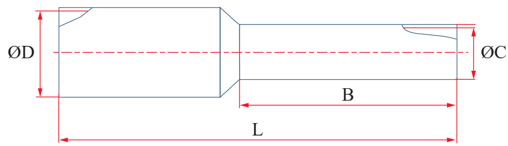
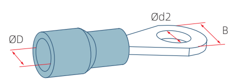


Table 3-5 Lug models and dimensions

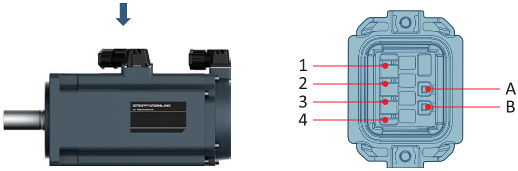
Lug model	L	B	ØC	ØD	Color
E0508	14 mm	8 mm	1.0 mm	2.6 mm	Orange
E1008	14 mm	8 mm	1.4 mm	3.0 mm	Yellow
E1508	14 mm	8 mm	1.7 mm	3.5 mm	Red

Table 3-6 Dimensions and appearance of TVR2-4 cable lugs of the grounding cable

Lug model		D	d2	B	Appearance
TVR	2-4	4.5mm	4.3mm	8.5mm	

3.5 Motor Connection

Terminal-type motor



Pin No.	Usage
1	Phase V
2	Phase U
3	Phase W
4	Grounding cable
A	Brake (polarity insensitive)
B	Brake (polarity insensitive)

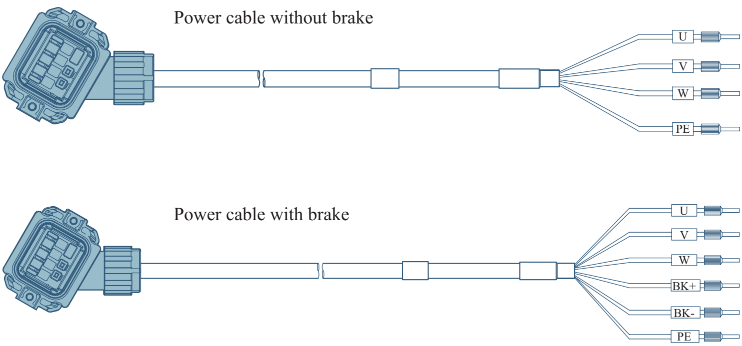
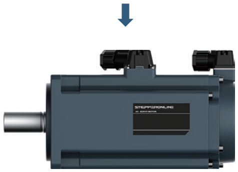
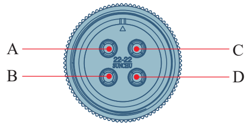


Figure 3-3 Power cable for terminal-type motor

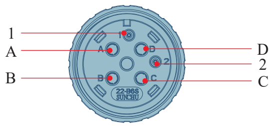
Aviation plug-type motor

Power input connector





Pin No.	Usage
A	Phase U
B	Phase V
C	Phase W
D	Grounding cable



Pin No.	Usage
A	Phase U
B	Phase V
C	Phase W
D	Grounding cable
1	Brake (polarity insensitive)
2	Brake (polarity insensitive)

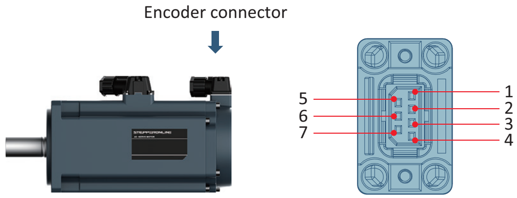
3.6 Encoder Connection (CN2)

CAUTION

Precautions for wiring of encoder signal cables:

- Ground the shield on the drive side and the motor side. Otherwise, the drive will report false alarms.
- Do not connect cables to "reserved" terminals.
- When determining the length of the encoder cable, take into full account the voltage drop caused by cable resistance and signal attenuation caused by distributed capacitance. Use shielded twisted pairs above 26 AWG (as per UL2464 standard) and keep the length within 10 m.

Terminal-type motor



Pin No.	Usage
1	DATA+
2	DATA-
3	BAT+
4	BAT-
5	+5V
6	0V
7	FG

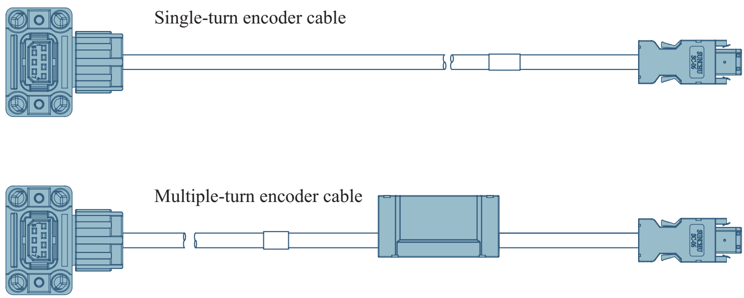


Figure 3-4 Absolute encoder signal cables

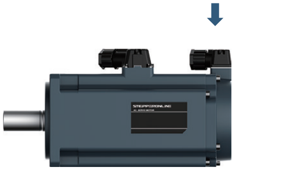
⚠ CAUTION

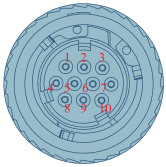
Battery box precautions:

- Install the battery in the correct direction. Do not pinch the connector cable when closing the battery box cover.
- Do not disassemble the battery because the internal electrolyte may spread out and cause physical injury.
- Do not short circuit the battery. Failure to comply may deteriorate the battery power and even incur the risk of explosion due to violent overheating.
- Before discarding the battery, insulate the battery with tape and then dispose of it according to local regulations.

Aviation plug-type motor

Encoder connector





Pin No.	Usage
1	DATA+
2	DATA-
4	+5V
5	BAT-
6	BAT+
9	0V
10	Enclosure

3.7 Control Signal Connection (CN1)

Control terminal	Connector Kit/Material No.	AWG
CN1	DB44	24~30

Use shielded signal cables to protect I/O signal circuits against strong interference noise at the periphery.

- Use a separate shielded cable for each type of analog signal.
- Shielded twisted pairs are recommended as digital signal cables.

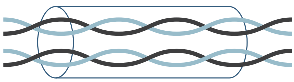


Figure 3-5 Shielded twisted pairs

CAUTION

- To avoid electromagnetic interference, keep a distance of at least 30 cm between I/O signal cables and power cables (input RST cables, output UVW cables, DC bus, and braking cables).

3.7.1 Position reference input signal

The pulse reference and symbol signal output circuit on the host controller side can either be differential drive output or open-collector output.

The following table lists the maximum input frequency and minimum pulse width of these output modes.

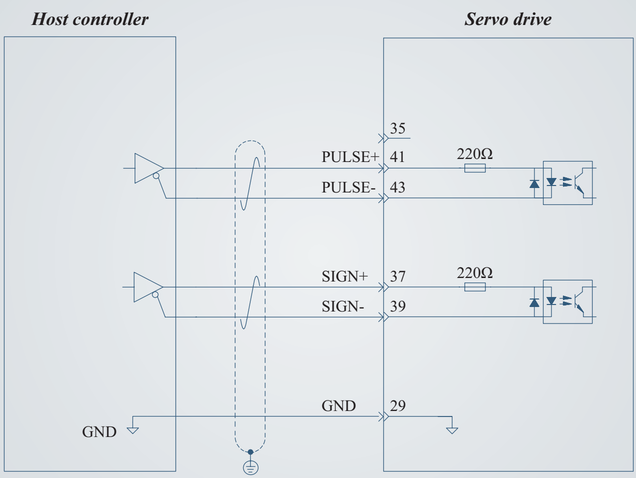
Pulse Mode		Single Channel Maximum Pulse Frequency (pps)	Minimum Pulse Width (µs)
Low speed	Differential	500k	1
	Open collector	200k	2.5
High-speed differential		4M	0.125

CAUTION

- The high-speed pulse and low-speed pulse cannot be used simultaneously, and only one of them can be used at a time.
- If the output pulse width of the host controller is smaller than the minimum value, an error occurs when the servo drive receives pulses.

Low-speed pulse reference input

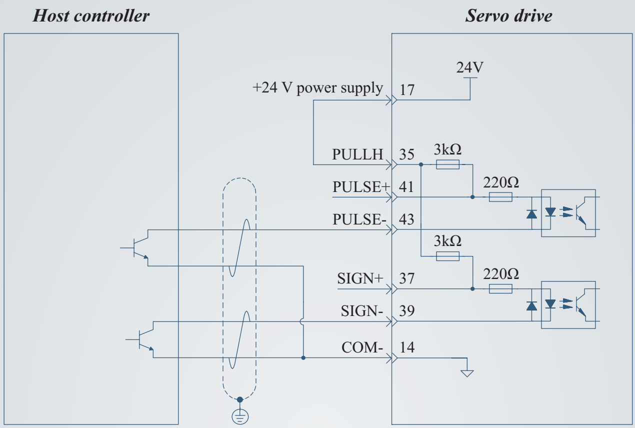
Differential mode

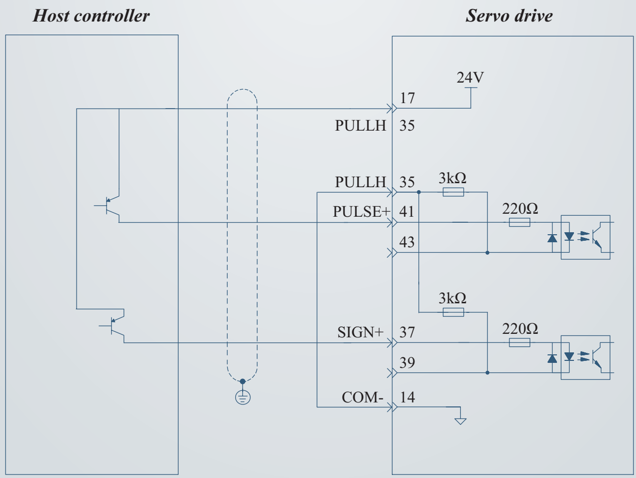


Note: This is a 5 V system. Please do not input the 24 V power.

Open collector mode

①When the internal 24 V power supply of the servo drive is used:

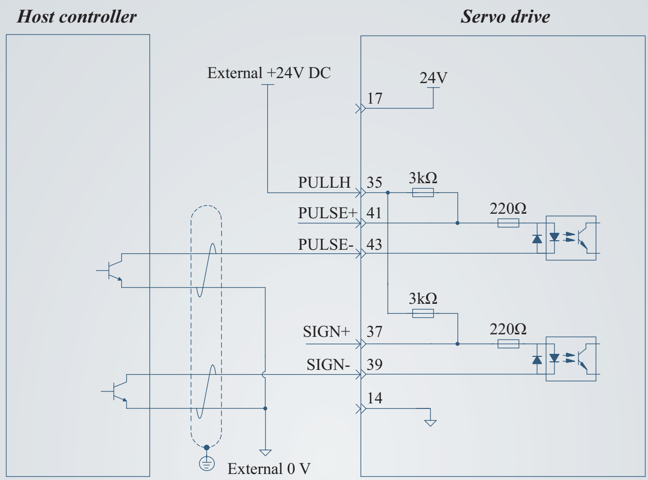


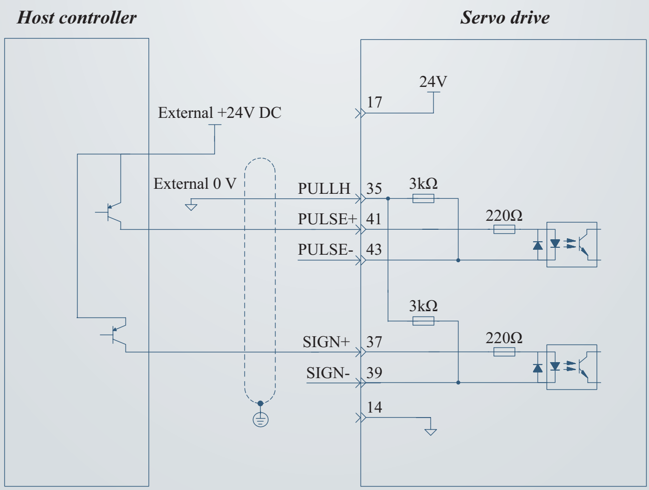


Open collector mode

②When the external power supply is used:

Scheme 1: using the internal resistor of the servo drive

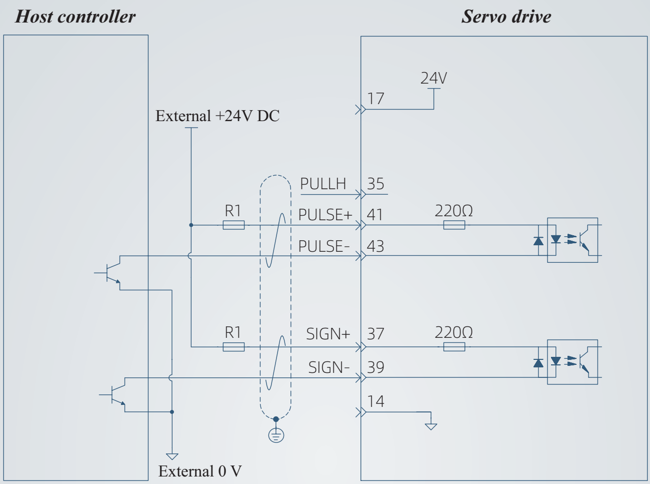


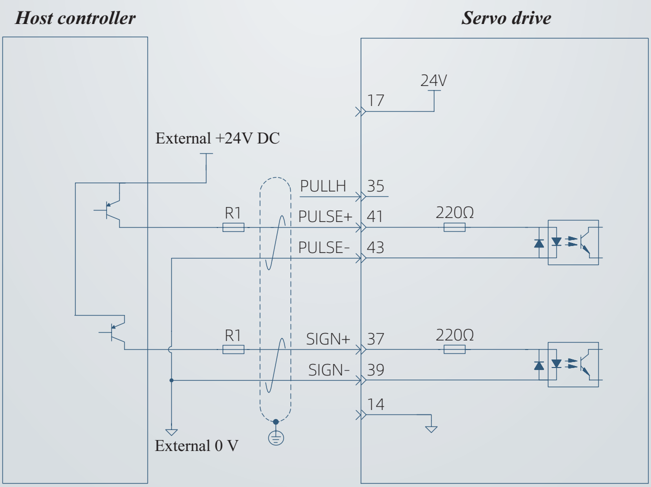


Open collector mode

②When the external power supply is used:

Scheme 2: using the external resistor





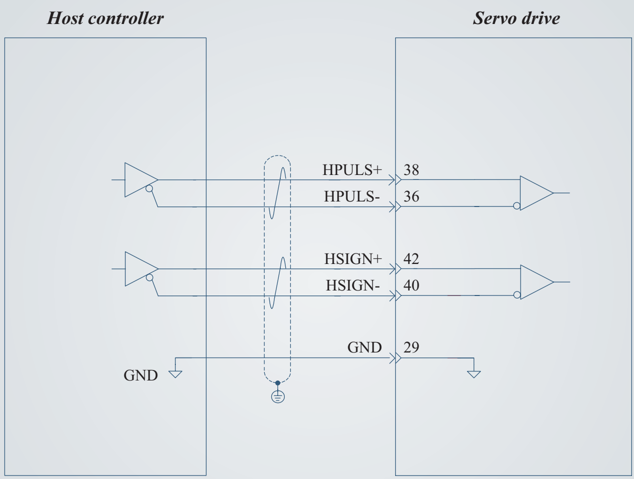
CAUTION

Avoid the following wiring errors:

- The current-limit resistor is not connected, resulting in burnout of terminals.
- Multiple terminals share the same current-limit resistor, resulting in the pulses receiving error.
- SIGN terminals are not connected, resulting in that these two terminals receive no pulses.
- Terminals are not correctly connected, resulting in burnout of terminals.
- Multiple terminals share the same current-limit resistor, resulting in the pulses receiving error.

High-speed pulse reference input

High-speed reference pulse and symbol signals on the host controller side can only be output to the servo drive via differential drive output.



Note: This is a 5 V system. Please do not input the 24 V power.

3.7.2 Analog input and output signals

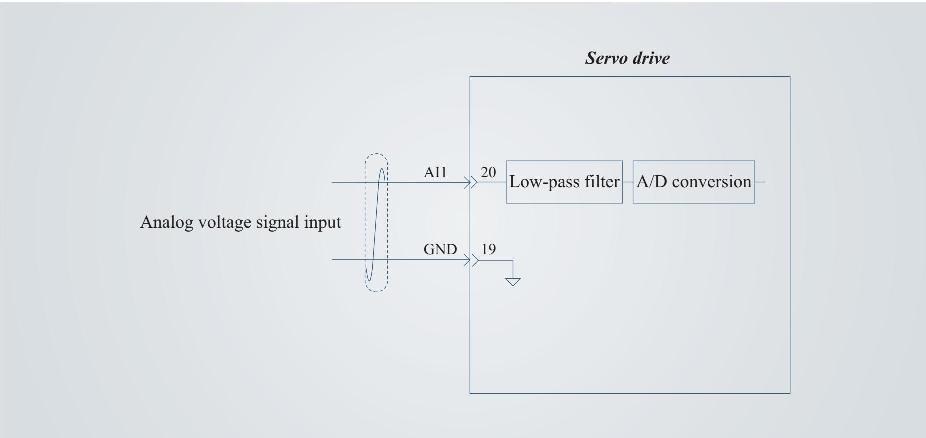
Analog input circuit

2-channel speed and torque analog signal input port: AI1/AI2

- AI1 voltage analog input (12-bit resolution)
  Input specification: -10 V to +10 V

Maximum allowable voltage: ±12 V
Input impedance: about 10 kΩ

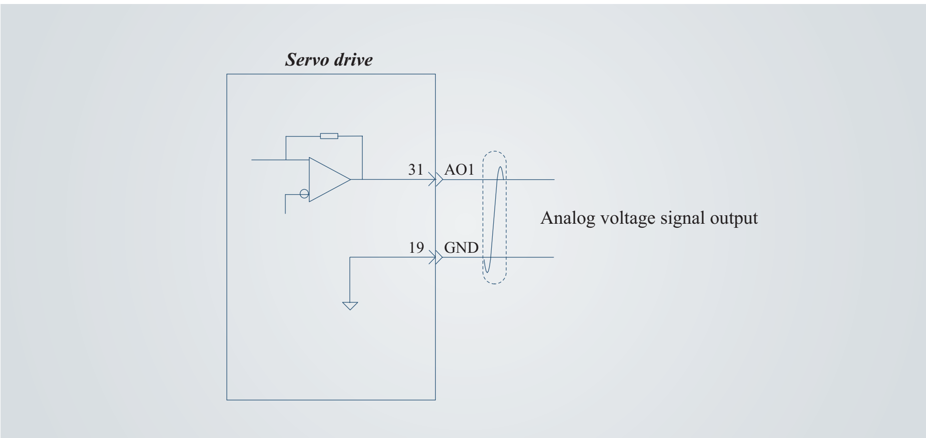
- AI2 current analog input
  Input specification: 4 mA to 20 mA
  Resolution: 12 bit



Analog output circuit

The speed and torque analog signal output terminal is AO1.

- Voltage range: -10 V to +10 V
- Output current limit: <1 mA



3.7.3 Digital input/output (DI/DO) signals

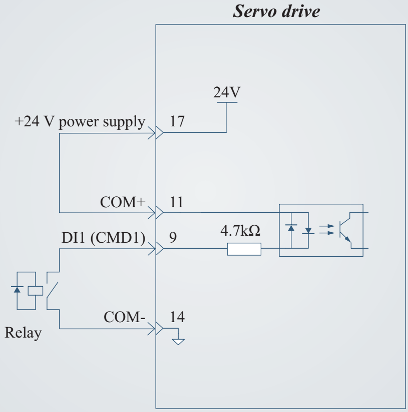
DI circuits

NOTICE

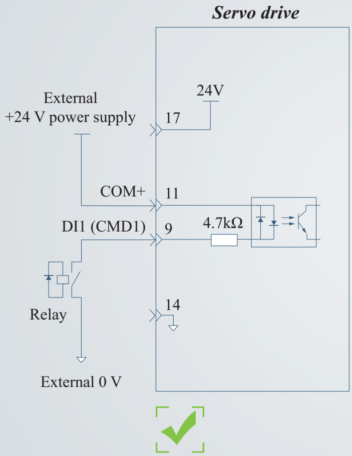
- The following takes DI1 as an example, and the remaining DI circuits are the same.

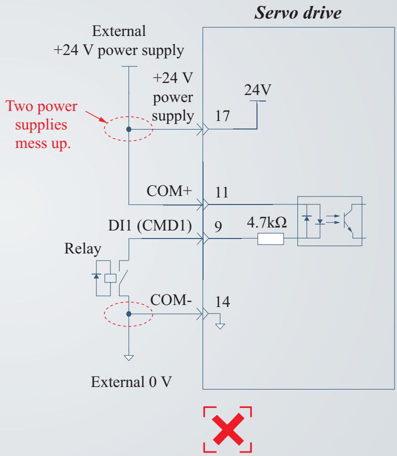
When the host controller adopts relay output:

- For use of the internal 24 V power supply of the servo drive.



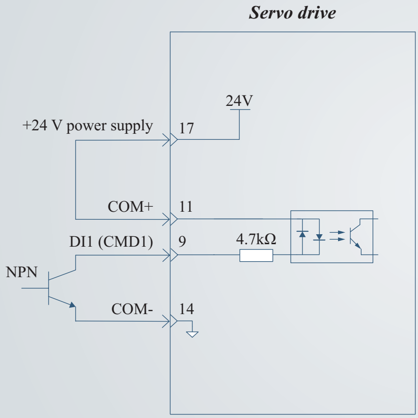
- For use of an external power supply.

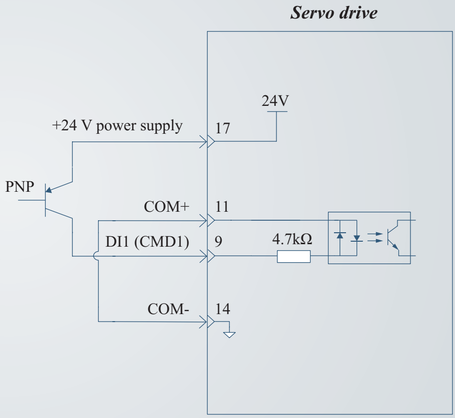




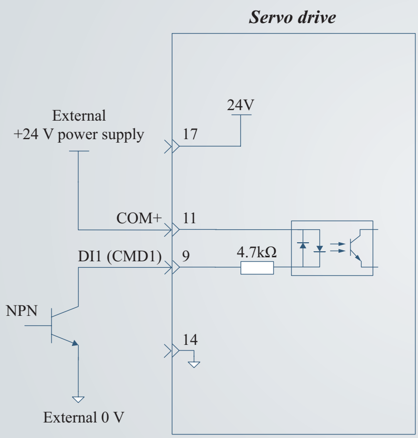
When the host controller adopts open collector output:

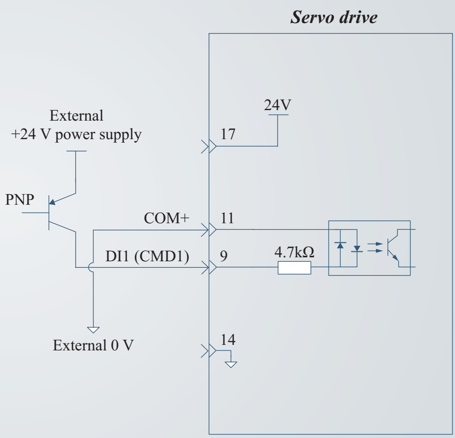
- For use of the internal 24 V power supply of the servo drive.





- For use of an external power supply.





NOTICE

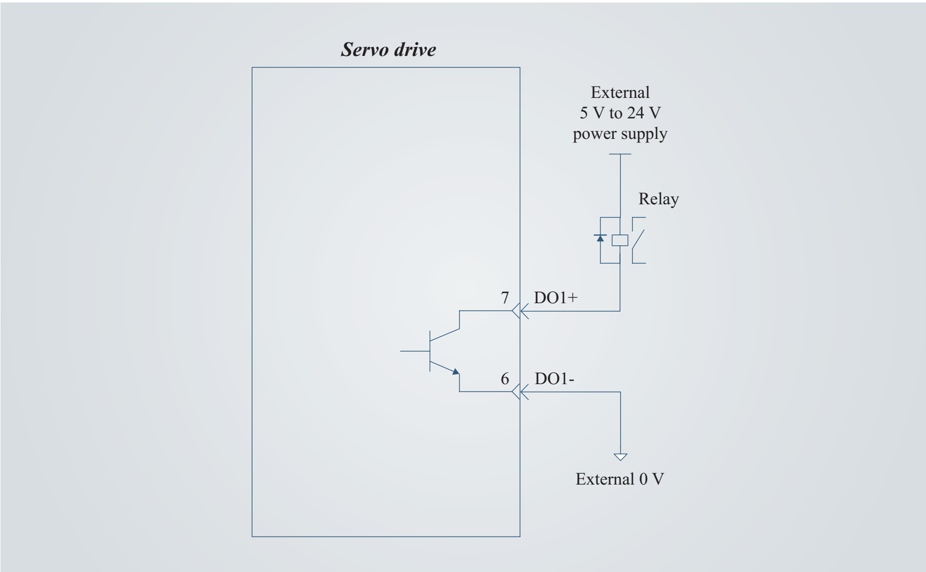
- PNP and NPN inputs cannot be used together in the same circuit.

DO circuits

NOTICE

- The following takes DO1 as an example, and the remaining DO circuits are the same.

When the host controller adopts relay input:



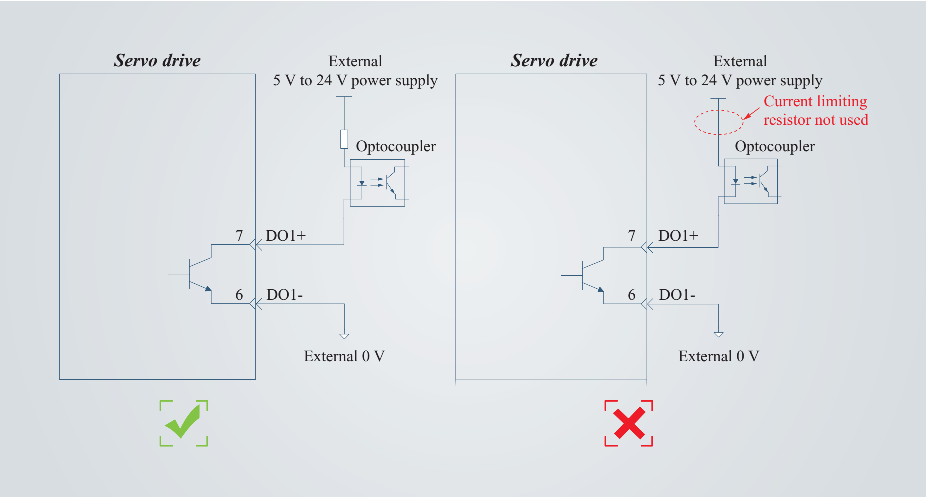
CAUTION

- When the host controller adopts relay input, a flywheel diode must be installed. Otherwise, the DO terminals may be damaged.

When the host controller adopts optocoupler input:

NOTICE

- The maximum allowable voltage and maximum current capacity of the optocoupler output circuit inside the servo drive are 30 VDC and DC50 mA, respectively.



3.7.4 Wiring of the brake

Some servo motors have a brake inside. The motor brake is used to prevent the movement of the servo motor shaft and keep the motor locked in the position when the servo motor may move unexpectedly due to external forces or its own weight during non-running conditions.

CAUTION

- The motor brake can only be used on a stopped motor and is only used to keep the load stationary. Do not use it to brake a moving load.

NOTICE

- The brake coil has no polarity.
- Switch off the S-ON signal after the servo motor stops.
- When the motor with a built-in brake runs, the brake may generate a click sound, which does not affect its function.
- When brake coils are energized (the brake is released), flux leakage may occur on the shaft end. Pay special attention when using magnetic sensors near the motor.

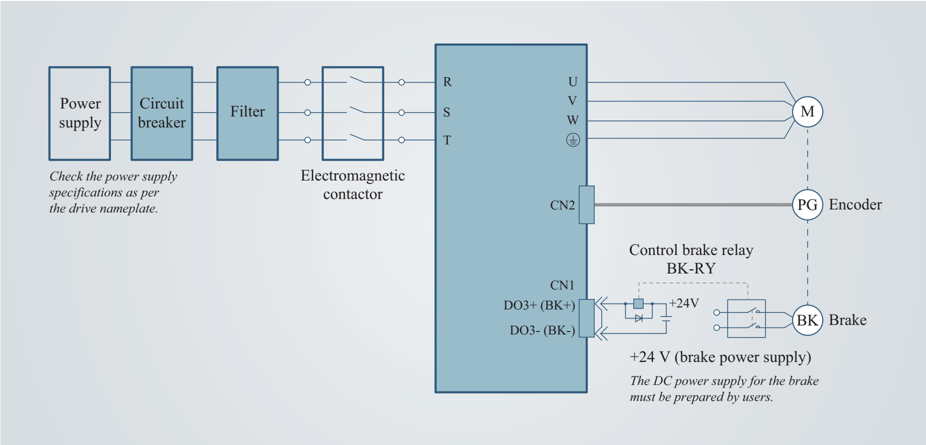


Figure 3-6 Wiring of the brake

Precautions during wiring:

When determining the length of the motor brake cable, take into full account the voltage drop caused by cable resistance. The input voltage must be at least 21.6 V to enable the brake to work properly.

Table 3-7 Brake specifications

Motor Model	Holding Torque	Rated Power	Power Supply Voltage
A6M40-100L2B1-M17	0.32N.m	6.9 W	24 V DC
A6M60-400H2B1-M17	1.27 N·m	7.3 W	24 V DC
A6M80-750H2B1-M17	3.2 N·m	8.5 W	24 V DC
A6M80-1000H2B1-M17	3.2 N·m	8.5 W	24 V DC

3.8 Communication Signal Connection (CN3 and CN4)

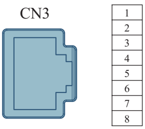
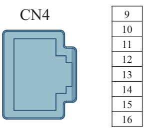
CN3 and CN4 are two same communication signal terminal connectors connected in parallel. Through CN3 and CN4, the following communication connections can be realized:

- Communication connection between the drive and PLC;
- Communication connection between drives.

NOTICE

RS485 communication connection of parallel drivers:

- When the number of nodes is large, the daisy chain connection mode is recommended for the RS485 bus.
- The RS485 signal reference ground of all nodes must be connected together. A maximum of 128 nodes can be connected.

CN3/CN4	Pin No.	Signal	Description
 CN3 communication terminals	4	RS485+	Data transmit+
	5	RS485-	Data transmit–
	6	-	-
	7	-	-
	8	GND	Data receive–
	Enclosure	PE	Shielding layer
 CN4 communication terminals	12	RS485+	Data transmit+
	13	RS485-	Data transmit–
	14	-	-
	15	-	-
	16	GND	Data receive–
	Enclosure	PE	Shielding layer

3.9 Communication Terminal Connection (CN6)

You can connect the drive to the PC through the CN6 terminal by using a serial cable (two-part wiring: Type-c to serial, and serial to USB) or a USB cable.

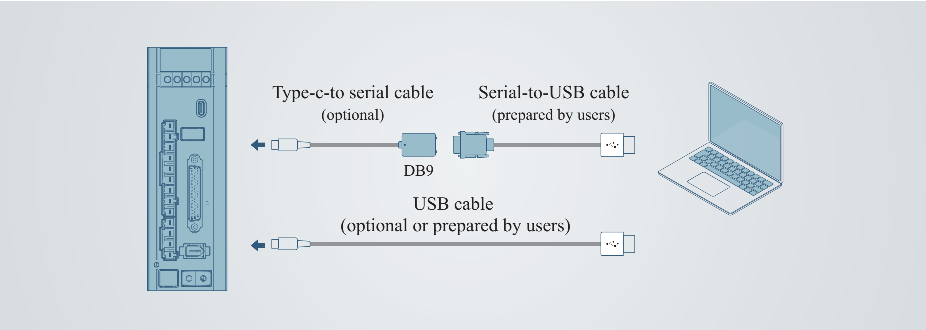
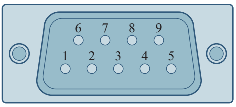


Figure 3-7 Communication terminal connection

DB9 Female Connector (Hole Type)	Pin No.	Signal	Description
	2	RXD	PC receive end
	3	TXD	PC transmit end
	5	GND	Ground
	Housing	PE	Shield

3.10 Braking Resistor Connection

CAUTION

- Do not connect the external braking resistor directly to positive and negative electrodes of the bus. Failure to comply may result in drive damage or even a fire.
- Do not select a resistor with resistance lower than the allowable minimum value. Failure to comply may result in an alarm or damage to the servo drive.
- Do not contact the external braking resistor during use. Failure to comply may result in burns caused by the hot braking resistor.

- Set parameters of the braking resistor properly before using the servo drive.
- Install the external braking resistor on incombustible objects (such as metal).

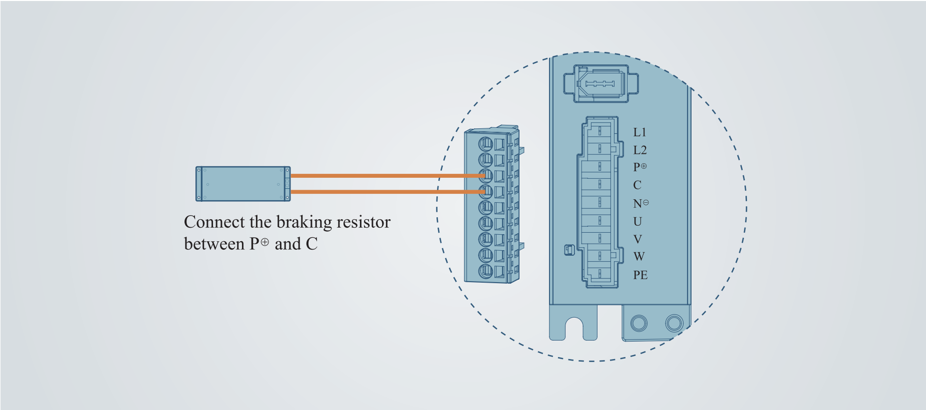


Figure 3-8 Connection of the external braking resistor

NOTICE

- The above connection example is for model SIZE A. Remove the jumper bar between terminals P⊕ and D before connecting an external braking resistor for other models.

Table 3-8 Specifications of the braking resistor

Drive Model		Resistance	Resistor Power	Minimum Allowable Resistance of External Resistor	Maximum Braking Energy Absorbed by Capacitor	Braking Resistor Type
SIZE A	A6-200RS	-	-	45 Ω	9.3 J	External braking resistor
	A6-400RS	-	-	45 Ω	26.29 J	External braking resistor
SIZE B	A6-750RS	50 Ω	50 W	40 Ω	22.41 J	Built-in braking resistor
SIZE C	A6-1000RS	25 Ω	80 W	20 Ω	26.70 J	Built-in braking resistor